

- Q.25 Describe the various factors affecting the selection of a rail route.
- Q.26 Write short note on lighting of tunnels.
- Q.27 Write the disadvantages of RUB.
- Q.28 Write the differences between a bridge and a culvert.
- Q.29 Write any five functions of ballast used in railway track construction.
- Q.30 Describe routine maintenance of bridges.
- Q.31 Describe the various functions of a pier provided in a bridge.
- Q.32 Describe the various advantages of providing the uniform gauge system of railway track in a country.
- Q.33 Describe the various functions of bridge bearings.
- Q.34 Describe double headed rails with the help of diagram.
- Q.35 Write a short note on inspection of bridges.

(1720)

No. of Printed Pages : 4

Roll No.

Subject:- Railways, Bridges and Tunnels

M.M. : 100

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- a) 50 lux c) 100 lux
b) 75 lux d) 125 lux

- | | |
|--------------------|-----------------|
| a) Grade separator | c) Minor Bridge |
| b) Major Bridge | d) Viaduct |

- a) 1767 mm c) 1777 mm
b) 1676 mm d) 1666 mm

- a) Abutments c) Columns
b) Piers d) Foundations

- a) Silt c) Clay
b) Sand d) Gravel

(1) 180753/030753/743

- Q.6 Which method is commonly used for drainage of tunnel carrying double railway track or two lane roads?
- a) Open Drain c) Central Drain
b) Side Drain d) Temporary Drain
- Q.7 The device allow free expansion, contraction and deflection of super structure of bridge is _____
- a) Foundations c) Columns
b) Bearing d) Piers
- Q.8 Which tunnel section is commonly used in traffic tunnel, subways or navigation tunnels?
- a) Horse-shoe shape c) Egg-Shaped
b) D-shape d) Elliptical Shape
- Q.9 The span for which, total cost of the bridge structure in minimum, is called _____
- a) Clear span b) Clearance
c) Economic span d) All of these
- Q.10 Which of the following is not a sub-structural part of a bridge _____
- a) Flooring c) Wing Walls
b) Abutment d) Piers

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 _____ is the frictional resistance of the surrounding soil on the outer surface of steining of wells.(Axial load/Skin friction)
- Q.12 The art to keep the tunnels free from water during and after and after the construction operations is called _____ (water logging / Drainage of tunnels)

- Q.13 For deciding the _____ of well foundation grip length must be considered as per IRC recommendations.
- Q.14 The railways provided over the ground surface are known as _____ (Elevated railways/Surface railways)
- Q.15 Fixed bearings do not permits the _____ movement of girders.
- Q.16 _____ of tunneling is used to construct tunnels in rocks. (Drift Method / Tramline Method)
- Q.17 The distance between the inner faces of a pair of wheels is called the _____
- Q.18 _____ is the operation of loading the rock, earth and any other excavated material.(Scaling / Mucking)
- Q.19 _____ is provided to distribute the load uniformly from sleepers to large area of formation
- Q.20 The way of placing the rails to an inward slope of 1 in 20 is termed as _____ (Tilting of rails / Coning of wheels)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write a short note on IRC classification of bridges.
- Q.22 Write any five functions of ballast used in railway track construction.
- Q.23 Write the problems arise during the sinking of a well foundation.
- Q.24 Describe routine maintenance of bridges.